Resources:
https://docs.google.com/spreadsheets/d/1UdDXniAo3598-IkF6toiSnNEB7vLfs1H41qPJ3NpTrA/edit?usp=sharing

Analysis of Wireshark Log Dump:

ATTACK TYPE:

DoS attack from Ip address 203.0.113.0

EVIDENCE:
At 3.390692 (Event 52) the first SYN request was sent to 192.0.2.1 from 203.0.113.0 over TCP. Which was met by a SYN ACK at 3.441926 (Event 53). This is normal behavior. However two SYN packets were sent at 3.664863 (Event 57) and 3.795332 (Event 59) before a SYN ACK could be sent back at 4.176295 (Event 64). After that the server was flooded with SYN requests from 203.0.113.0 to port 443 and 9 SYN packets were sent to the server before the server could respond with a reset acknowledgement (RST, ACK) packet at event 83.

After this interaction the server was flooded with SYN packets with RST ACK responses at events 89, 95, 100, 103, 109, 117, and 124. This flood of SYN packets overloaded the server and the only traffic after event 121 was from or to 203.0.113.0.

EFFECTS:
Denial of service to any users attempting to connect after 20.167744 (Event 122)

RESOLUTION STEPS:
Add a firewall filtering rule to disallow any traffic from 203.0.113.0
Configure firewall rate limiting for SYN requests
Review firewall logs to monitor for any similar traffic